TEST CODE **01207030**

FORM TP 2007003

JANUARY 2007

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

BIOLOGY

Paper 03 – General Proficiency

1 hour

15 JANUARY 2007 (p.m.)

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. **In addition to the 1 hour, candidates are allowed a reading time of 10 minutes. Writing may begin during the 10-minute period.**
2. **This paper is divided into THREE sections. Answer THREE questions, ONE from Section A, ONE from Section B and ONE from Section C.**
3. **Where appropriate, answers should be illustrated by diagrams.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

Copyright © 2005 Caribbean Examinations Council®.
All rights reserved.

SECTION A

Answer ONE question from this section.

1. (a) (i) Briefly explain how different pH levels are produced along the length of the alimentary canal in humans.

(ii) Give TWO reasons why the elongated tubular shape of the alimentary canal is suited to its digestive function. **(8 marks)**

(b) Following a healthy lifestyle with a balanced diet is considered important in weight control in humans.

(i) Identify FOUR constituents of a balanced diet.

(ii) What are the likely effects of age and lifestyle on the constituents of a balanced diet of an individual? **(8 marks)**

(c) In certain cases, an obese individual undergoes stomach stapling surgery in which the volume of the stomach is reduced.

(i) Explain why this is an option to control the diet.

(ii) Give TWO disadvantages of this process for the individual. **(4 marks)**

Total 20 marks

2. (a) (i) Briefly explain how nitrogen from the air is cycled through living systems.

(ii) Suggest TWO reasons why the cycling process in (a)(i) is important. **(8 marks)**

(b) Human activities sometimes disrupt natural cycles that are important to living organisms.

(i) Identify TWO human activities that disrupt natural cycles or processes.

(ii) For EACH activity identified at (b)(i), describe the precautions that may be taken to reduce disruption of the natural cycles. **(6 marks)**

(c) The spread of many diseases can be reduced by human activity such as hygienic (sanitary) and lifestyle practices.

(i) Identify TWO diseases that occur because of poor hygiene and inappropriate lifestyle practices.

(ii) Explain how proper hygiene and sound lifestyle practices can help reduce the occurrence of the diseases you identified at (c)(i). **(6 marks)**

Total 20 marks

SECTION B

Answer ONE question from this section.

3. (a) (i) What are the differences in structure between a mature red blood cell and a white blood cell?

(ii) Explain TWO likely effects of a relatively small number of red blood cells on the body. **(6 marks)**

(b) (i) Trevor cuts his finger and his mother wraps it with a piece of cloth to stop the flow of blood.

Suggest TWO advantages and TWO disadvantages of his mother's action.

(ii) Often our immediate response to symptoms of the common cold is to reach for an antibiotic. What may be the disadvantages of this response? **(8 marks)**

(c) (i) Identify TWO types of immunity and explain how they provide protection for the body.

(ii) Despite extensive medical research, viral diseases such as AIDS have been particularly difficult to cure.

What may be the benefits of encouraging pregnant women to be tested for the AIDS virus? **(6 marks)**

Total 20 marks

4. (a) Using examples, distinguish between receptors and effectors. **(4 marks)**

(b) While walking to school one day, Vinita sees a ferocious dog about to attack. She turns and runs.

(i) Explain how the eye and the central nervous system function to cause Vinita to see and respond to the dog.

(ii) The hormone adrenalin is produced immediately. Explain the changes that take place in Vinita's body as a result of the production of adrenalin. **(8 marks)**

(c) (i) Describe how the movement of muscles and bones in the leg occurs for Vinita to raise her knee and run.

(ii) After a while Vinita's legs become tired and she is forced to rest.

Explain why her leg muscles become tired and what happens to them when she rests. **(8 marks)**

Total 20 marks

GO ON TO THE NEXT PAGE

SECTION C

Answer ONE question from this section.

5. (a) With the aid of a diagram, describe THREE ways in which the xylem is suited to its function. **(8 marks)**

(b) (i) One agricultural practice is that of watering plants.

Explain why too much water is dangerous to plants.

(ii) Another agricultural practice observed in the Caribbean is that of covering newly transplanted seedlings with coconut branches.

Suggest why a farmer may adopt this practice.

(iii) A few weeks after transplanting, it is the practice to remove some healthy seedlings while allowing others to grow.

Explain why this process becomes necessary. **(12 marks)**

Total 20 marks

6. (a) Using a diagram of a biomass pyramid, describe the relationships between the trophic levels in a named habitat. **(8 marks)**

(b) (i) Suggest TWO advantages and TWO disadvantages of rearing fish in an aquarium.

(ii) In some countries, aquaculture (breeding fish in freshwater ponds) is being promoted as a means of increasing fish production.

Explain why a fish species whose natural habitat is the sea should NOT be bred in freshwater ponds. **(8 marks)**

(c) Sometimes waterways become polluted by oil spills. Suggest how this may affect marine organisms. **(4 marks)**

Total 20 marks

END OF TEST